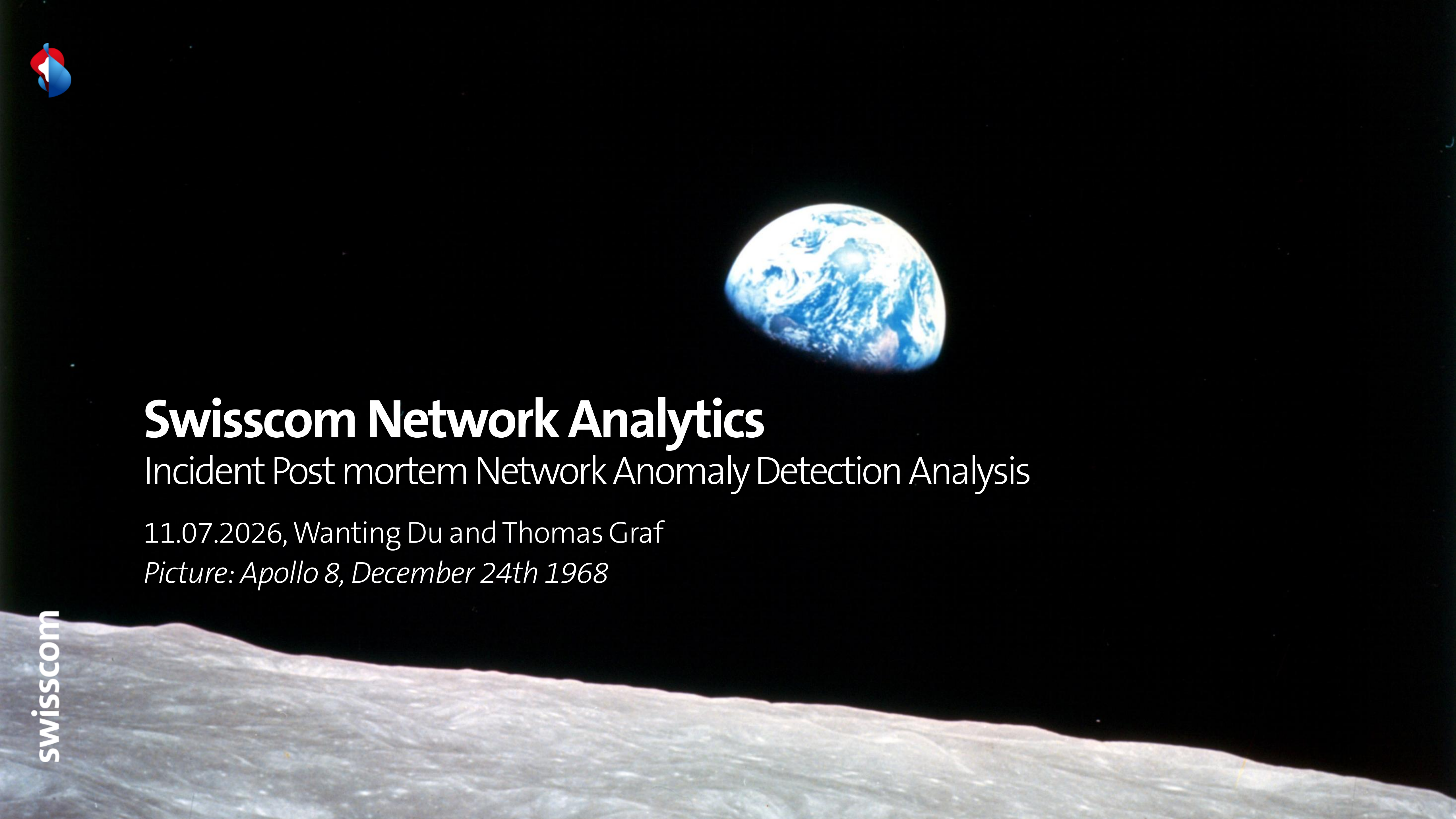


Swisscom Network Analytics

Incident Post mortem Network Anomaly Detection Analysis

11.07.2026, Wanting Du and Thomas Graf

Picture: Apollo 8, December 24th 1968

July 2nd, BGP Routing Loop causing high CPU

Network Anomaly detected on 42 L3 VPN's excessive BGP path withdrawals

On July 2nd at 10:02, a new VRF was introduced at two Inter-AS Option A peering nodes. **This configuration change triggered on 42 L3 VPN's across Swisscom BGP routing Loops. The BGP process CPU usage increased on all nodes sharing the same routing table. Cosmos Bright Lights Network Anomaly Detection observed the excessive BGP route withdrawal and alerted.**

The network observability team analyzed at 16:18 the Network Anomaly Detection activity and deducted that no packet drops nor missing traffic, but excessive BGP update, and withdrawal activities were observed and that the origin from the excessive BGP activity is **originated from the Inter-AS Option A route-distinguishers at 16:40** and contacted at 17:05 network operation control.

At 17:36 the contact to the responsible network operation team was established. **A maintenance window was identified as the triggering event** and network incident filed. Since no connectivity service impact was identified during the network frozen zone, the resolvement of the network incident has been postponed to business hours next day.

On July 3rd at 13:13, the involved **Inter-AS Option A peerings where shutdown and the BGP loop was terminated.**

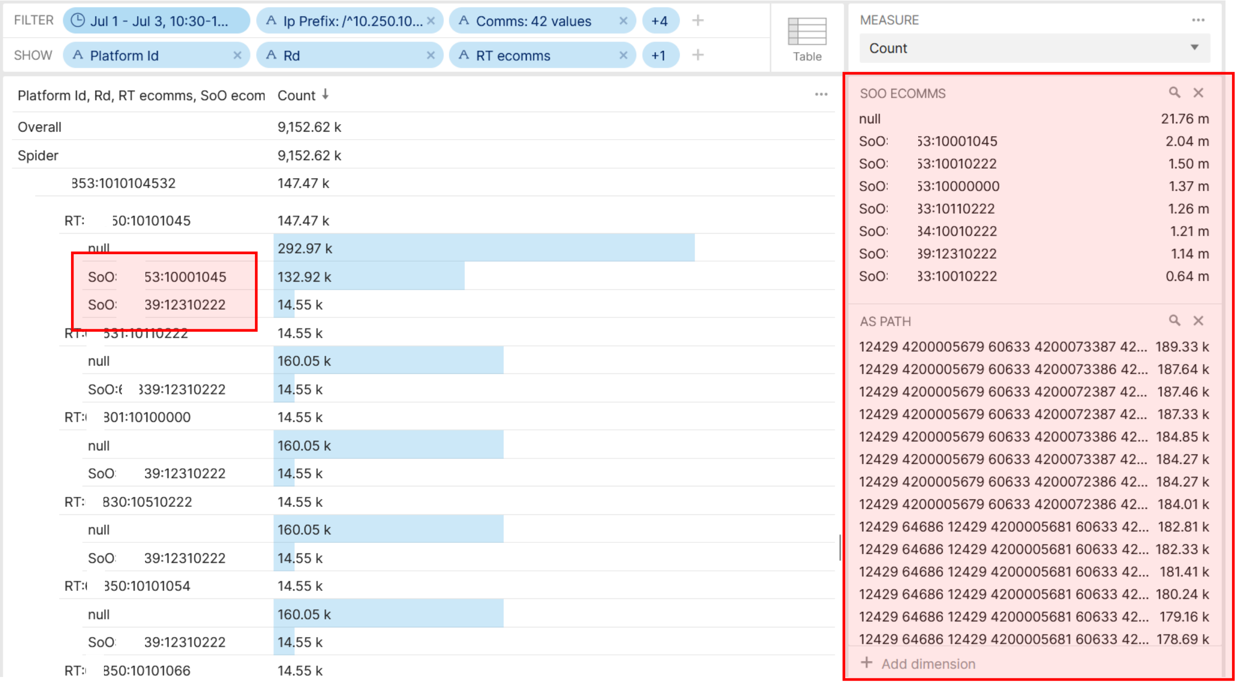
July 2nd, BGP Routing Loop causing high CPU

Loop Details

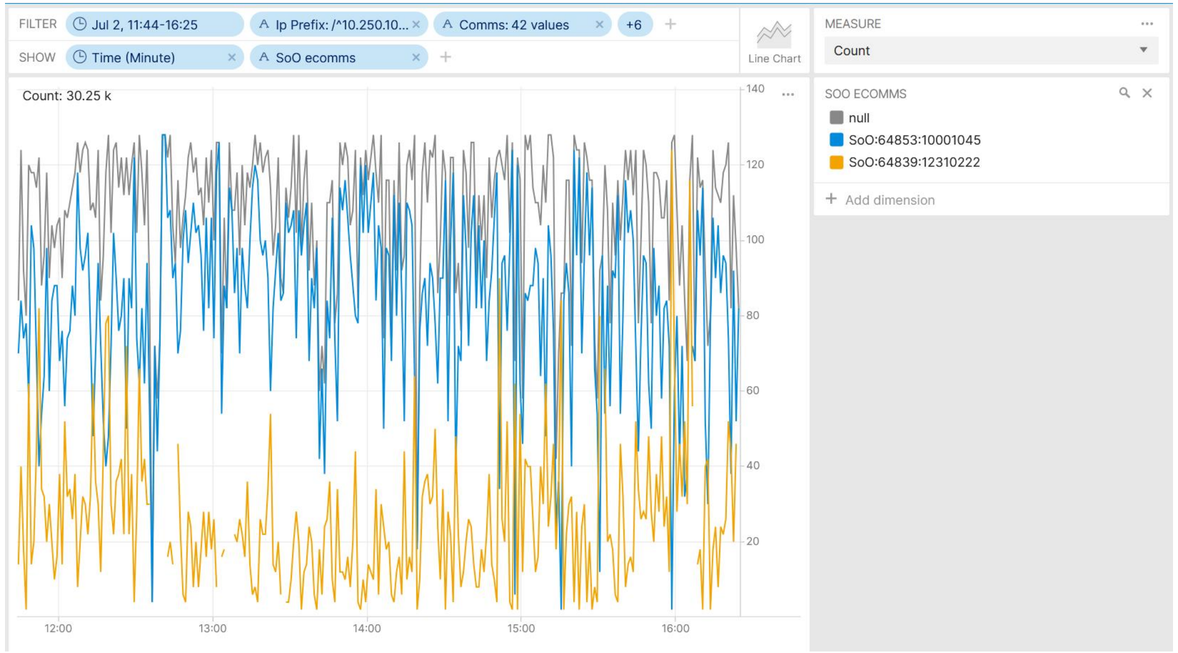
https://datatracker.ietf.org/doc/html/rfc4364#section-7

How PEs Learn Routes from CEs

Every CE whose site is not in a transit VPN can use the same ASN. This can be chosen from the private ASN space, and it will be stripped out by the PE. Routing loops are prevented by use of the Site of Origin attribute



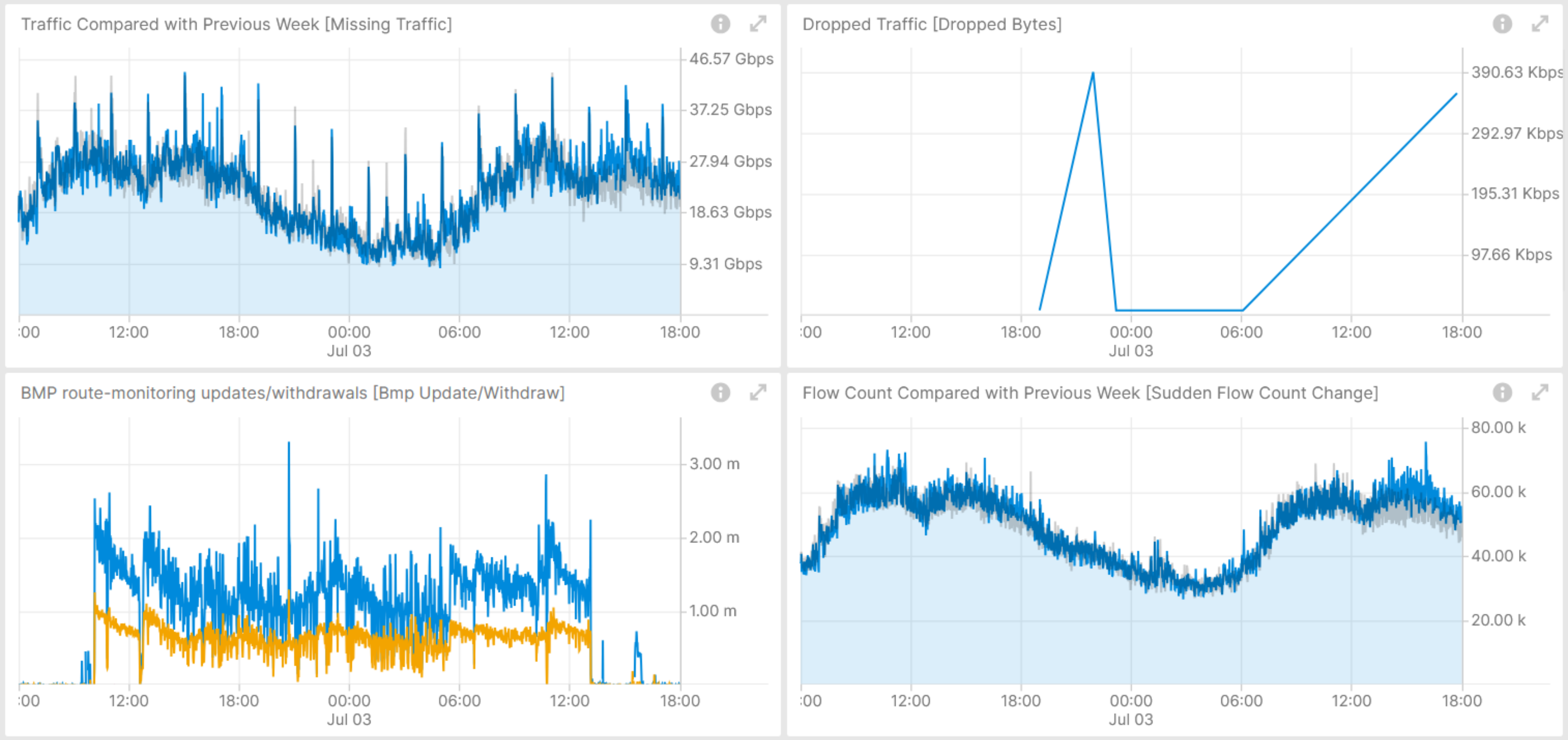
Operational Network Telemetry BMP collected metrics showing on which L3 VPN shares multiple Site of Origin Attributes



Operational Network Telemetry BMP collected metrics showing on a particular L3 VPN how the Site of Origin Attributes oscillating

July 2nd, BGP Routing Loop causing high CPU

L3 VPN Operational Network Telemetry Metrics



Operational Network Telemetry forwarding plane, IPFIX, BMP measured control plane metrics

Shows no drops and unrelated decreased traffic volume to previous week, **Measured with IPFIX and Correlated with with BGP VPNv4/6, BMP Adj-RIB In and Local RIB.**

Shows excessive BGP topology changes and no flow count changes. **Measured with IPFIX and and Correlated with BGP VPNv4/6, BMP Adj-RIB In and Local RIB.**

July 2nd, BGP Routing Loop causing high CPU

L3 VPN Network Anomaly Relevant State Notification

Network Anomaly Detection system monitoring x:44080
L3 VPN in real-time during maintenance window

Concern Score: 37
Flow Count Spike: **18**
Missing Traffic: **19**
Traffic Drop: **0**
BMP Peer/Interface Down: **99/7**
BMP Update/Withdrawal: **99/100**

BMP route-monitoring Update/Withdraw check recognized the excessive withdrawals.

BMP peer Down/Up check detected the BGP peer state change, and the concern object point to the mgtboro1010zhb and mgtboro1010zhh nodes and peers.

Interface Down/Up check did not apply.

Traffic Drop spike did not apply since dropped packets where not observed.

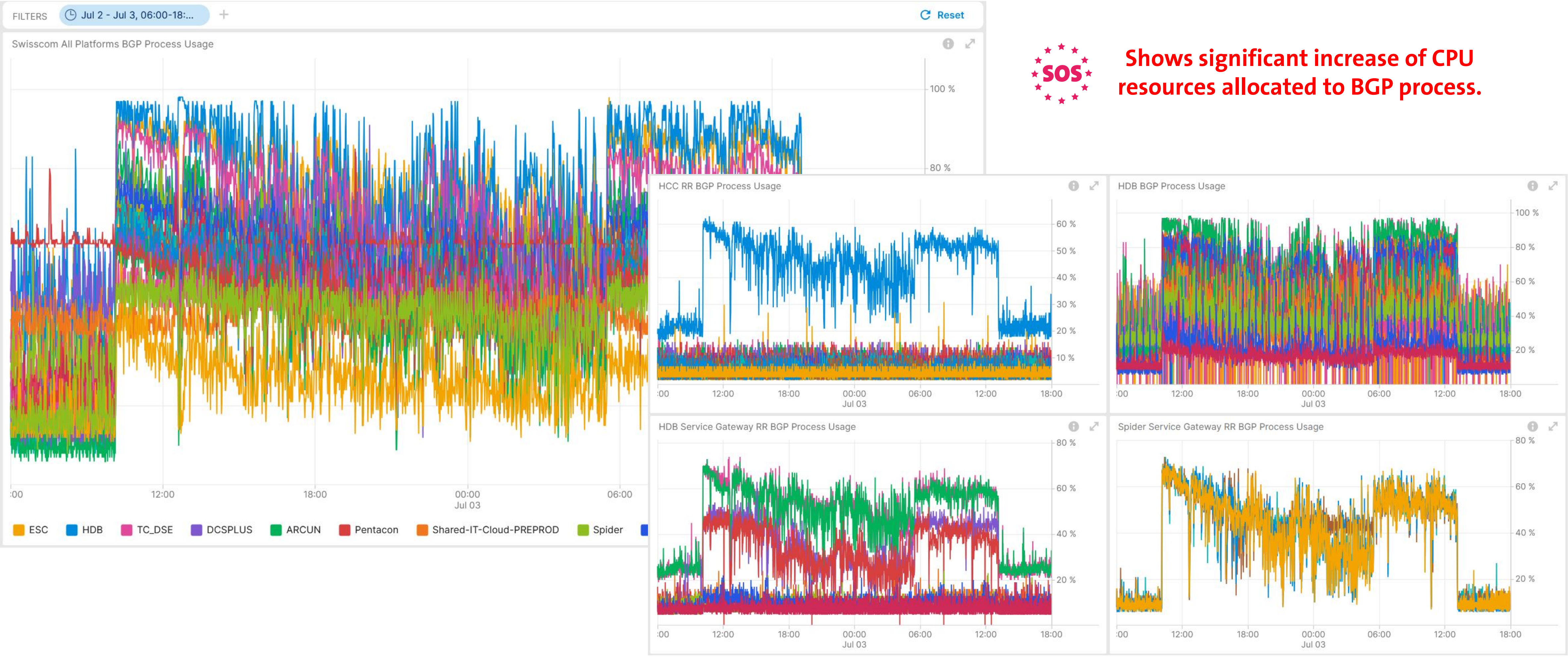
Missing Traffic did not apply.

Increased or decreased flow count changes did not apply.

Overall: 3 out of 6 checks have detected the topology changes without forwarding plane impact.

July 2nd, BGP Routing Loop causing high CPU

CPU Usage Impact Analysis



Shows significant increase of CPU resources allocated to BGP process.

Operational Network Telemetry YANG-Push collected metrics

July 2nd, BGP Routing Loop causing high CPU

BGP Prefix and Peering Detail Analysis

Shows BGP prefixes and which Inter-AS Option peers involved in routing loop.

IP Prefixes causing BGP routing Loops (00:30775)

Ip Prefix	Count ↓
103.71/32	166,684.51 k
103.68/32	163,569.56 k
103.69/32	163,251.78 k
103.67/32	163,158.13 k
103.70/32	162,778.66 k
103.66/32	161,816.70 k
103.65/32	160,430.12 k
103.8/32	160,242.78 k
103.64/32	160,192.20 k
103.1/32	160,069.20 k
103.7/32	159,831.12 k
103.2/32	159,517.15 k
103.5/32	159,109.61 k
103.3/32	158,819.14 k
103.6/32	158,756.45 k
103.4/32	158,446.97 k
2.91/32	2,441.67 k
47.0/24	436.92 k

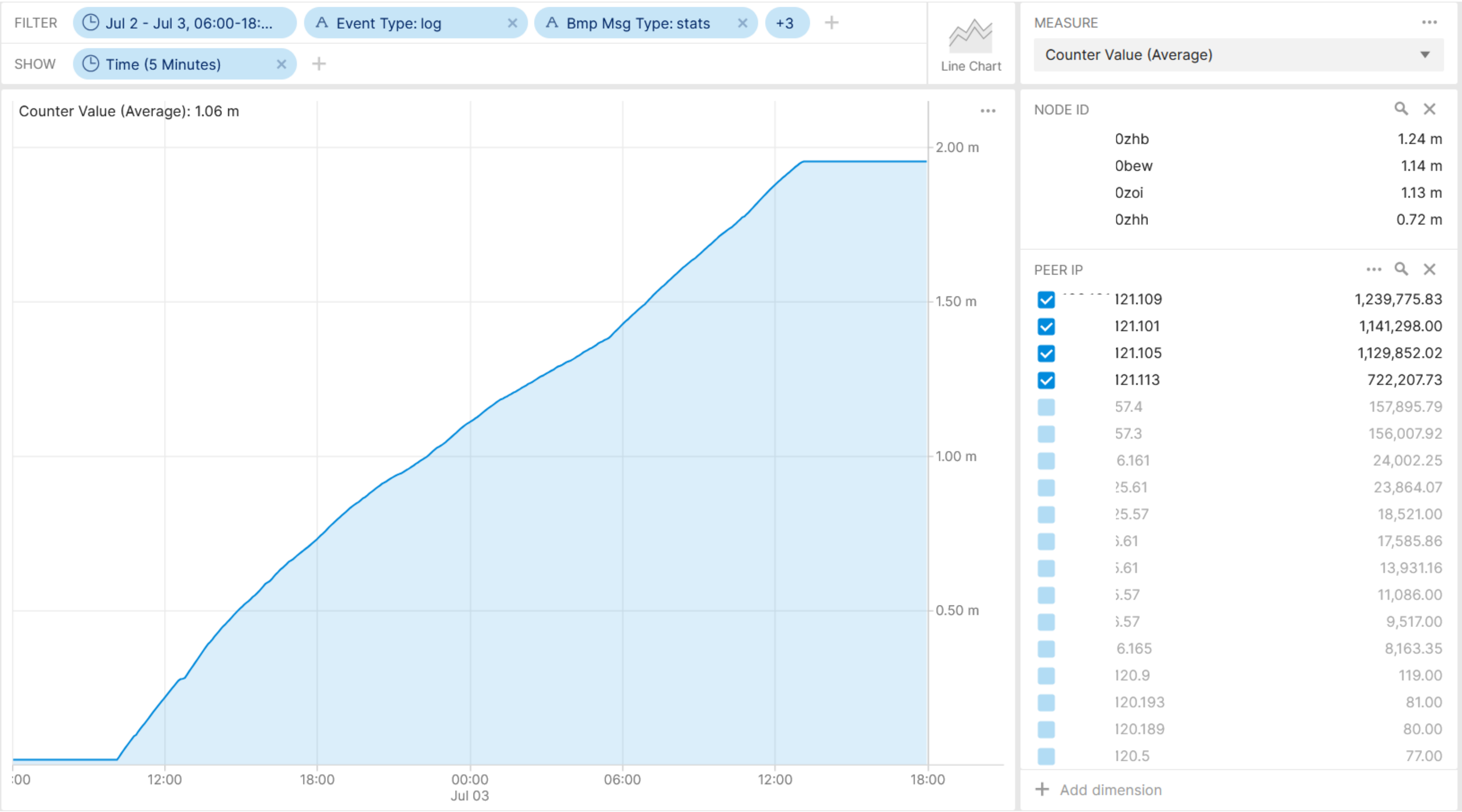
Service Gateway Inter-AS Option A Peering Receiving Looping Prefixes

Platform Id, Node Id, Rd, Peer Ip	Count ↓
	1,436.11 k
20bew	420.79 k
005712:9	420.79 k
0.101.121.101	247.42 k
0.0	173.37 k
20zhb	387.65 k
005714:8	387.65 k
0.101.121.109	216.30 k
0.0	171.35 k
20zoi	367.65 k
005713:9	367.65 k
0.101.121.105	219.45 k
0.0	148.20 k
20zhh	260.03 k
005715:8	260.03 k
0.0	140.01 k
0.101.121.113	120.02 k

Operational Network Telemetry BMP collected metrics

July 2nd, BGP Routing Loop causing high CPU

BMP Peering statistics Duplicated Analysis



Operational Network Telemetry BMP collected metrics

Shows that BGP duplicated prefix advertisement was detected on Inter-AS Option A BGP peerings.

Stat Type = 1: (32-bit Counter) Number of (known) duplicate prefix advertisements

https://datatracker.ietf.org/doc/html/rfc7854#section-4.8

July 2nd, BGP Routing Loop causing high CPU

Generative AI **Expert Operational** Analysis

SpeakToDaisy Agent

Message SpeakToDaisy Agent

Thomas Graf

Analyze BGP process CPU (`process-cpu-one-minute`) from `daisy.prod.device-metric-proc-cpu-utilization` correlated with BMP route updates from `daisy.prod.control-bmp` for (`comms`) [6 0775] between July 2nd 06:00 and July 3rd 18:00. Group by (`platform_id`) and hour. Show top 10 impacted platforms with avg/max CPU %, baseline comparison, and severity classification.

BGP Process CPU Analysis with BMP Route Update Correlation

Based on the analysis of BGP process CPU utilization from `daisy.prod.device-metric-proc-cpu-utilization` correlated with BMP route updates from `daisy.prod.control-bmp` for community `0775` between **July 2nd 06:00 and July 3rd 18:00, 2026**, here are the findings:

Top 10 Impacted Platforms by BGP CPU Utilization

Rank	Platform	Avg CPU %	Max CPU %	Baseline Avg	CPU Increase	Severity	Peak Hour
1		7.58%	36%	0.33%	**+2,197%**	🔴 CRITICAL	09:00 (11.54%)
2		6.59%	77%	0.88%	**+648%**	🔴 CRITICAL	09:00 (11.75%)
3		6.28%	64%	0.34%	**+1,747%**	🔴 CRITICAL	09:00 (10.88%)
4		5.59%	72%	0.30%	**+1,763%**	🟠 HIGH	09:00 (10.39%)
5		5.13%	60%	0.18%	**+2,750%**	🟠 HIGH	09:00 (9.56%)
6		0.02%	7%	0.01%	+56%	🟢 NORMAL	-

Thomas Graf

what were the looped prefixes?

🔴 **Looped Prefixes Identified**

Primary Affected Range 3.0/24

The following `/32` prefixes within this range show definitive routing loop characteristics:

Prefix	Total Updates	Unique Paths	Avg Path Length	Loop Severity
71/32	166.7M	564	**89.3 hops**	🔴 CRITICAL
68/32	163.6M	576	**83.4 hops**	🔴 CRITICAL
69/32	163.3M	564	**89.3 hops**	🔴 CRITICAL
67/32	163.2M	563	**89.4 hops**	🔴 CRITICAL
70/32	162.8M	571	**83.3 hops**	🔴 CRITICAL
66/32	161.8M	573	**83.4 hops**	🔴 CRITICAL
65/32	160.4M	561	**89.4 hops**	🔴 CRITICAL
8/32	160.2M	561	**89.4 hops**	🔴 CRITICAL
64/32	160.2M	575	**83.4 hops**	🔴 CRITICAL
1/32	160.1M	576	**83.4 hops**	🔴 CRITICAL
7/32	159.8M	573	**83.4 hops**	🔴 CRITICAL
2/32	159.5M	564	**89.4 hops**	🔴 CRITICAL
5/32	159.1M	575	**83.5 hops**	🔴 CRITICAL
3/32	158.8M	576	**83.4 hops**	🔴 CRITICAL

🔍 Evidence of Routing Loops

1. Abnormally Long AS Paths

- **Normal BGP paths**: 3-8 AS hops
- **Observed paths: 83-89 AS hops** (10-30x normal)
- This indicates routes circulating through the same ASes repeatedly

2. Massive Update Volume

- **16.6 billion total updates** across the 0.103.x range
- **~6 million updates per hour per prefix** during peak (09:00-11:00)
- Each prefix received continuous updates for **28+ hours**

3. Path Instability

- **560-576 unique AS paths** per prefix
- Constant path modifications indicate routes being re-advertised with incremented AS paths

4. Synchronized Timing

- **Start**: All .103.x prefixes began at **07:03:00 UTC, July 2nd**
- **End**: All ceased at **~11:13:31 UTC, July 3rd**
- This synchronized behavior suggests a common root cause

July 2nd, BGP Routing Loop causing high CPU
Generative AI **Novice Operational** Analysis

Thomas Graf
on July 2nd a bgp routing loop occured. Please list the prefixes involved in the routing loop

Prefixes Involved in the Routing Loop/Instability

The most affected prefixes showing signs of routing loop behavior (indicated by repeated AS numbers in the AS_PATH and extremely high update counts) are:

Primary Affected Prefixes (Loop Detection):

1. **)3.71/32** - 389M+ updates
2. **)3.67/32** - 383M+ updates
3. **)3.69/32** - 382M+ updates
4. **)3.70/32** - 382M+ updates
5. **)3.68/32** - 381M+ updates
6. **)3.66/32** - 380M+ updates
7. **)3.1/32** - 379M+ updates
8. **)3.65/32** - 378M+ updates
9. **)3.8/32** - 376M+ updates
10. **)3.64/32** - 375M+ updates

Evidence of Routing Loop:

The AS_PATH analysis reveals clear loop patterns with repeated AS numbers:

- **AS 12429** appears repeatedly (up to 12 times consecutively) in paths like:
  - `6837 65123 12429 12429 12429 12429 12429 12429 12429 12429 12429 12429`
- Complex loop patterns involving multiple ASes:
  - `6837 65123 12429 64655 12429 64686 12429 4200005681 60633 4200072386...`

Key Observations:

- **Total BGP updates on July 2nd**: ~7 billion events (4.9B updates + 2.1B withdrawals)
- **Loop indicators**: AS_PATH prepending anomalies with AS 12429 appearing multiple times consecutively
- **Affected ASes**: 6837, 65123, 12429, 64655, 64686, 4200005681, 60633, and various private ASNs (4200072386, 4200072387, 4200073386)

This pattern strongly suggests a **BGP routing loop** involving the 10.250.103.0/24 subnet, with AS 12429 being a key participant in the loop. The massive number of updates (hundreds of millions per prefix) indicates route flapping consistent with loop behavior.

Thomas Graf
for those prefixes, which comms in the BGP AS 64497 do they have in common?

Key Observations:

1. **Two distinct community groups** from AS 64497:
   - **High-frequency group**: `':11347` and `7:6` (~1B updates each)
   - **Medium-frequency group**: `':10316`, `:19257`, `7:511`, `7:5591` (~381M updates each)
2. **Additional communities** predominantly from AS 60633 were also present on all prefixes:
   - `3:100`, `3:1111`, `:217`, `:209`, `3:1001`, `6` `1032`, etc.
3. **Extended Communities** observed:
   - `RT:` `)000113` (Route Target)
   - `RT:` `)014579` (Route Target)

Thomas Graf
for prefixes with comms :6 and I347, which ecomms starting with SoO (site-of-origin) do they have in common?

Based on my analysis of the BGP telemetry data for prefixes with communities **97:6** and **97:11347**, here are the **Site-of-Origin (SoO) extended communities** that are common across all 10 affected prefixes:

Common SoO Extended Communities

SoO Extended Community	Update Count	Prefix Count	Significance
SoO: 0:30775	511,692,702	10/10	Primary SoO identifier
SoO: 9:1	219,220,272	10/10	Secondary SoO identifier
SoO: 9:16718	13,985,669	10/10	Tertiary SoO identifier

Thomas Graf
can you confirm that these comms where affected?
316
347
657
3947
1
591
338
339
7845
4080
5813

Yes, I can **confirm that ALL 11 communities you listed were affected** during the July 2nd routing loop event. Here's the detailed breakdown:

Confirmation of Affected Communities

Community	Total Updates	Prefix Count	Status
347	1,039,817,958	10/10	✅ **AFFECTED**
)316	381,454,746	10/10	✅ **AFFECTED**
I1	381,454,746	10/10	✅ **AFFECTED**
591	381,454,746	10/10	✅ **AFFECTED**
4080	743,666,965	10/10	✅ **AFFECTED**
338	381,454,746	10/10	✅ **AFFECTED**
339	381,454,746	10/10	✅ **AFFECTED**
7845	381,454,746	10/10	✅ **AFFECTED**
3657	65,866,412	10/10	✅ **AFFECTED**
8947	65,866,412	10/10	✅ **AFFECTED**
5813	12,240,539	10/10	✅ **AFFECTED**

July 2nd, BGP Routing Loop causing high CPU

Generative AI **Expert Analytical** Analysis

Thomas Graf

search in `daisy.prod.bl-streaming-alerts-ietf-flattened` which `vpn id` has `Max Anomaly Bmp Withdraw Symptom Concern Score` higher than `99` for time period July 2nd 06:00 AM and July 3rd and 06:00 PM and list them. Summarize when the score reached 100 and when it went back to normal. Find common `Anomaly BMP Peer Withdraw` dimensions.

Based on my analysis of the `daisy.prod.bl-streaming-alerts-ietf-flattened` datasource for the time period July 2nd 06:00 AM to July 3rd 06:00 PM, here are the findings:

VPN IDs with Max Anomaly BMP Withdraw Symptom Concern Score > 99

I identified **13 VPN IDs** that experienced concern scores reaching 100 during this period:

VPN ID	Max Score	First Occurrence	Last Occurrence
7	100	2026-07-02 06:04 AM	2026-07-03 11:47 AM
16	100	2026-07-02 08:26 AM	2026-07-03 11:09 AM
47	100	2026-07-02 08:27 AM	2026-07-03 11:10 AM
57	100	2026-07-02 08:31 AM	2026-07-02 08:58 AM
947	100	2026-07-02 08:26 AM	2026-07-03 11:09 AM
	100	2026-07-02 08:26 AM	2026-07-03 11:09 AM
1	100	2026-07-02 08:26 AM	2026-07-03 11:09 AM
1	100	2026-07-02 08:26 AM	2026-07-03 11:09 AM
8	100	2026-07-02 08:26 AM	2026-07-03 11:09 AM
9	100	2026-07-02 08:26 AM	2026-07-03 11:09 AM
345	100	2026-07-02 08:26 AM	2026-07-03 11:09 AM
080	100	2026-07-02 08:27 AM	2026-07-02 11:23 AM
313	100	2026-07-02 08:27 AM	2026-07-03 10:56 AM

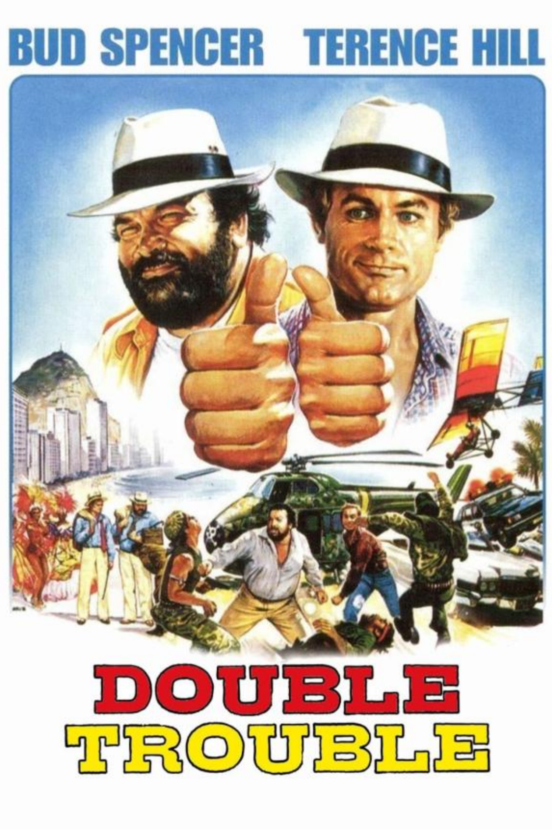
Timeline Summary: When Score Reached 100 and Returned to Normal

Escalation to Critical (Score = 100):

- **Start Time:** July 2nd, 2026 at **08:26 AM UTC**
- The anomaly concern score escalated from 99 to 100 at this time across most affected VPNs
- Prior to 08:26, scores were at 99 (starting around 08:13 AM)

Return to Normal:

- **End Time:** July 3rd, 2026 at **11:09-11:11 AM UTC**
- At 11:09 AM, the last critical events (score=100) were recorded
- By **11:11 AM**, the concern score dropped significantly to **63**, indicating the anomaly subsided
- This represents a **~27-hour incident window**



Common Dimensions in Anomaly BMP Peer Withdraw

Analysis of the Route Distinguisher (RD) dimensions reveals **strong commonality** across affected VPNs:

Most Frequently Affected Route Distinguishers (appearing across multiple VPNs):

1. `0:6` `21022213` - Present in ALL top 8 VPNs (highest occurrence: 2,506 events)
2. `0:6` `21022214` - Present in ALL top 8 VPNs (highest occurrence: 2,505 events)
3. `0:6` `21022210` - Present in ALL top 8 VPNs (highest occurrence: 2,289 events)
4. `0:6` `11022231` - Present in ALL top 8 VPNs (highest occurrence: 1,983 events)
5. `0:6` `71022213` - Present in ALL top 8 VPNs (highest occurrence: 1,940 events)
6. `0:6` `11022230` - Present in ALL top 8 VPNs (highest occurrence: 1,915 events)
7. `0:6` `71022212` - Present in ALL top 8 VPNs (highest occurrence: 1,900 events)

Recommendation:

This appears to be a **coordinated BGP withdrawal event** affecting multiple VPNs sharing common route distinguishers. The simultaneous impact across 8+ VPNs with identical RD patterns suggests:

- A common upstream peer or route reflector issue
- Potential maintenance or configuration change affecting shared infrastructure
- Possible route leak or withdrawal propagation from a common source

Would you like me to investigate any specific VPN or Route Distinguisher in more detail, or analyze the correlation with other anomaly types during this time period?

Thomas Graf

yes, analyze the correlation with other anomaly types during this time period

Timeline Correlation Analysis

sql Copy code

```sql
July 2, 2026:
06:00 ──┬── BMP Peer Down events begin (early warning for some VPNs)
        │
08:00 ──┼── BMP Withdraw CRITICAL events BEGIN (score=99→100)
        │   BMP Update events BEGIN (perfect correlation)
        │
08:13 ──┼── First critical withdraw event recorded
        │
08:26 ──┼── Peak withdraw activity begins
        │
12:00 ──┴── Some peer down events end (64497:511)

July 3, 2026:
03:25 ──┬── Interface status changes (64497:5591 only)
        │
11:09 ──┼── BMP Withdraw CRITICAL events END
        │   BMP Update events END
        │   Score drops to 63 by 11:11
        │
14:34 ──┴── Late BMP Peer Down events (64498:1338/1339)
```

IETF NMOP - Semantic Metadata Annotation for Network Anomaly Detection

draft-ietf-nmop-network-anomaly-lifecycle/semantics – **Schema Tree**

```
notifications:
   +---n relevant-state-notification
      +--ro publisher
      |  +--ro id?         yang:uuid
      |  +--ro name        string
      |  +--ro version?    string
      +--ro id                                  yang:uuid
      +--ro uri?                                inet:uri
      +--ro description?                        string
      +--ro start-time                          yang:date-and-time
      +--ro end-time?                           yang:date-and-time
      +--ro smcblsymptom:strategy?              string
      +--ro confidence-score?                   score
      +--ro concern-score                       score
      +--ro (service)?
      |  +--:(smtopology:l2vpn)
      |  |  +--ro smtopology:vpn-service* [vpn-id]
      |  |     +--ro smtopology:vpn-id                string
      |  |     +--ro smtopology:uri?                  inet:uri
      |  |     +--ro smtopology:vpn-name?             string
      |  |     +--ro smtopology:site-ids*             string
      |  |     +--ro smtopology:change-id?            yang:uuid
      |  |     +--ro smtopology:change-start-time?
      |  |     |       yang:date-and-time
      |  |     +--ro smtopology:change-end-time?
      |  |             yang:date-and-time
      |  +--:(smtopology:l3vpn)
      |     +--ro smtopology:vpn-service* [vpn-id]
      |        +--ro smtopology:vpn-id                string
      |        +--ro smtopology:uri?                  inet:uri
      |        +--ro smtopology:vpn-name?             string
      |        +--ro smtopology:site-ids*             string
      |        +--ro smtopology:change-id?            yang:uuid
      |        +--ro smtopology:change-start-time?
      |        |       yang:date-and-time
      |        +--ro smtopology:change-end-time?
      |                yang:date-and-time
```

```
notifications:
   +---n relevant-state-notification
   +--ro anomaly* [id revision]
         +--ro id                                        yang:uuid
         +--ro revision                                  yang:counter32
         +--ro uri?                                      inet:uri
         +--ro state                                     identityref
         +--ro description?                              string
         +--ro start-time
         |       yang:date-and-time
         +--ro end-time?
         |       yang:date-and-time
         +--ro confidence-score?                         score
         +--ro pattern?                                  identityref
         +--ro annotator
         |  +--ro id?              yang:uuid
         |  +--ro name             string
         |  +--ro version?         string
         |  +--ro annotator-type?  enumeration
         +--ro symptom!
         |  +--ro id                                   yang:uuid
         |  +--ro concern-score                        score
         |  +--ro smcblsymptom:action?                 string
         |  +--ro smcblsymptom:reason?                 string
         |  +--ro smcblsymptom:trigger?                string
         |  +--ro smcblsymptom:network-plane?          enumeration
         |  +--ro smcblsymptom:template?               string
         |  +--ro smcblsymptom:season?                 Enumeration
         +--ro smtopology:vpn-node-terminations*
                 [hostname route-distinguisher]
            +--ro smtopology:hostname                    inet:host
            +--ro smtopology:route-distinguisher         string
            +--ro smtopology:peer-ip*                    inet:ip-address
            +--ro smtopology:next-hop*                   inet:ip-address
            +--ro smtopology:interface-id*               uint32
```

Shows the observed symptoms, the network dimensions triggering and connectivity service impacted.

July 2nd, BGP Routing Loop causing high CPU

Problem Confirmed and Analyzed **Annotation**

```
module: ietf-relevant-state
  +--rw relevant-state
     +--rw id                                      yang:uuid
     +--rw uri?                                    inet:uri
     +--rw description?                            string
     +--rw start-time                              yang:date-and-time
     +--rw end-time?                               yang:date-and-time
     +--rw strategy?                               string
     +--rw confidence-score?                       score
     +--rw concern-score                           score
     +--rw stage                                   identityref
     +--rw (service)?
     |  +--:(anom-srv-topo:l3vpn)
     |     +--rw anom-srv-topo:l3vpn-service* [vpn-id]
     |        +--rw anom-srv-topo:vpn-id                  string
     |        +--rw anom-srv-topo:uri?                    inet:uri
     |        +--rw anom-srv-topo:vpn-name?               string
     |        +--rw anom-srv-topo:site-ids*               string
     |        +--rw anom-srv-topo:change-id?              yang:uuid
     |        +--rw anom-srv-topo:change-start-time?
     |        |       yang:date-and-time
     |        +--rw anom-srv-topo:change-end-time?
     |                yang:date-and-time
     +--rw anomaly* [id revision]
        +--rw id                                      yang:uuid
        +--rw revision                                yang:counter32
        +--rw uri?                                    inet:uri
        +--rw stage                                   identityref
        +--rw description?                            string
        +--rw start-time
        |       yang:date-and-time
        +--rw end-time?
        |       yang:date-and-time
        +--rw confidence-score?                       score
        +--rw pattern?                                identityref
        +--rw annotator
        |  +--rw id?                yang:uuid
        |  +--rw name               string
        |  +--rw version?           string
        |  +--rw annotator-type?    enumeration
        +--rw operational-data
        |  +--rw topic-name?      message-broker-topic
        |  +--rw subject-name?    message-broker-subject
        +--rw symptom!
        |  +--rw id                               yang:uuid
        |  +--rw concern-score                    score
        |  +--rw anom-symptom:action?             string
        |  +--rw anom-symptom:reason?             string
        |  +--rw anom-symptom:trigger?            string
        |  +--rw anom-symptom:network-plane?      enumeration
        |  +--rw anom-symptom:template?           string
        |  +--rw anom-symptom:season?             enumeration
        +--rw anom-srv-topo:vpn-node-terminations*
                [hostname vrf-name]
           +--rw anom-srv-topo:hostname                inet:host
           +--rw anom-srv-topo:vrf-id?                 uint32
           +--rw anom-srv-topo:vrf-name                string
           +--rw anom-srv-topo:route-distinguisher?    string
           +--rw anom-srv-topo:interface-id*           uint32
           +--rw anom-srv-topo:interface-name*         string
           +--rw anom-srv-topo:peer-ip*                inet:ip-address
           +--rw anom-srv-topo:next-hop*               inet:ip-address
```

annotate and tag

```
                        +-------------+
            +--------> |  Detection  | ---------+
Adjustments |           +-------------+          | Symptoms
            |                                    |
            |                                    v
   +------------+                        +------------+
   | Refinement |<---------------------- | Validation |
   +------------+        Problem         +------------+
                       Confirmation
```

INSA Lyon is working in collaboration with Swisscom on a Network Anomaly Detection Postmortem system proof of concept with the aim to facilitate statistical relevant-state data labeling.

Labels are chosen for an alert-id, where tags can be chosen or defined and shared among multiple alerts, and annotations are being defined for a single alert-id and time frame by a human in the loop.

Labels

operational-correctness: Correct
analytical-correctness: Correct

Tags

"BGP Routing Loop"

Annotations

```
{
  "Incident Number": "INC000012345",
  "reference": "https://oneitsm.example.com/example/inc=INC000012345"
},
{
  "Maintenance Window Number": "CRQ0000056789",
  "reference": "https://oneitsm.example.com/example/crq=CRQ0000056789"
}
```

Network Anomaly Detection, Postmortem and Generative AI

Key discoveries and how NMOP documents fit

Key Discoveries

- Large Lange Models understand operational network telemetry timeseries data.
- With RAG's (Retrieval-Augmented Gen) network and anomaly symptom semantics are given as external context to AI agents.
- YANG, YANG schema registry and streaming catalog enable RAG's.
- LLM's do have network knowledge. They know what a BGP routing loop is. They can subnet. They understand the relationship between BGP process CPU consumption and BGP routing loop.
- **The preliminary results proves that with the right operational and analytical network data, the right integration and some knowledge results are impressive.**

- draft-ietf-nmop-yang-message-broker-integration
  Makes YANG schemas available in YANG schema registry and streaming catalog so that humans and AI agents can discover which network data is discoverable where.
- draft-ietf-nmop-yang-message-broker-message-key
  Establishes a message broker topic naming and message indexing scheme to minimize data consumption. Enables topic compaction for current state data compression.
- draft-netana-nmop-message-broker-bmp-telemetry-msg
  Transforms BMP, BGP control-plane, network data in a standard fashion to YANG and JSON. YANG semantics enable AI agents to understand BMP meta and BGP data.
- draft-ietf-nmop-network-anomaly-semantics Defines human and AI agent understandable network anomaly symptom semantics.
- draft-ietf-nmop-network-anomaly-lifecycle Defines the network anomaly lifecycle process where AI assisted network perform data labeling in a postmortem system.

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